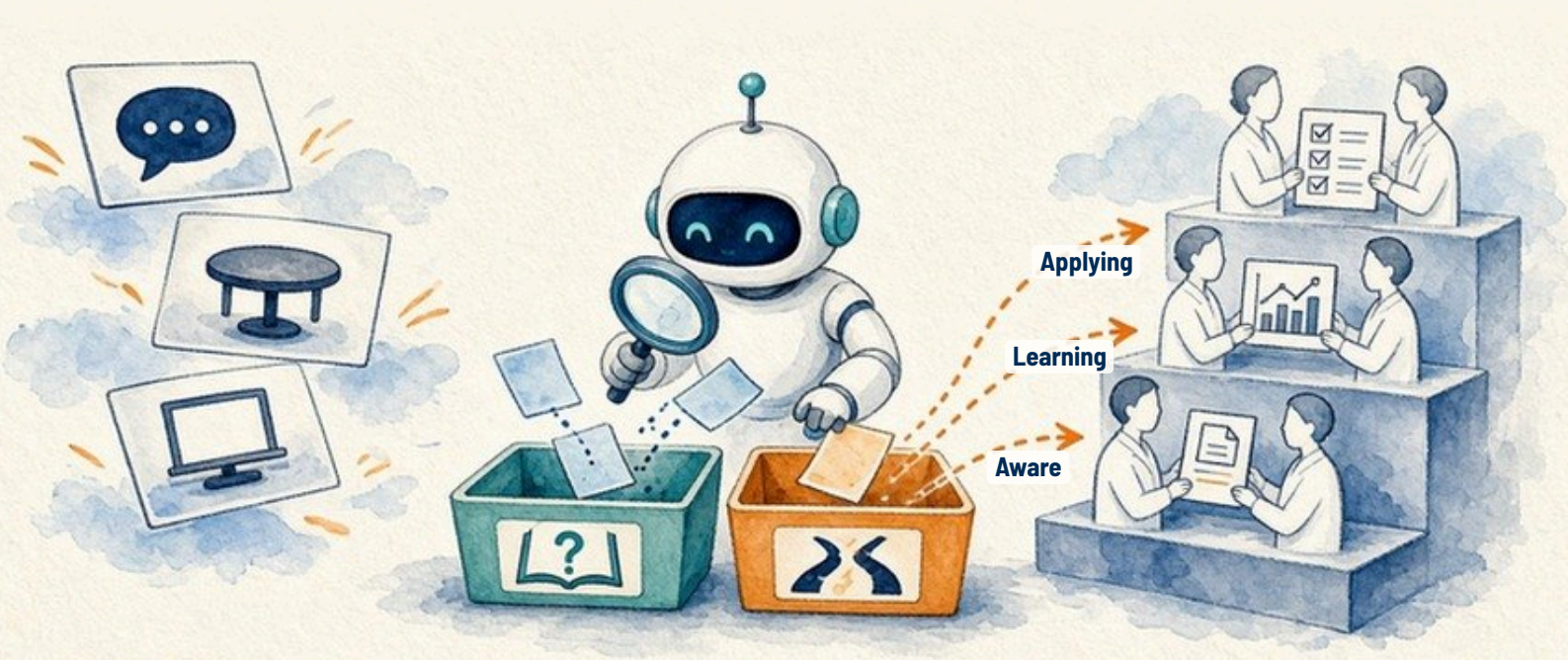


02

Insights Engine

Trace every care gap to the HCP knowledge or belief gap behind it, then match education to each HCP segment and stage.

Hackathon vertical 02 of 04 · build afternoon agenda, swarm, gates, question bank and impact worksheet

Facilitators: John and Anjali (insights pillar leads; they lead the education route: medical content and medical education)

AI in Action for Medical Affairs

October 13–14, 2026 · Convene Philadelphia (2nd floor), Two Commerce Square, 2001 Market St

Host and keynote: Vivek Mukhatyar · **aiinaction.up.railway.app/hackathon**

Scan for the hackathon pack

02 AT A GLANCE

The vertical on one page

THE CHALLENGE

Field insights, advisory boards and enquiries keep pointing at the same care gaps, but the HCP knowledge or belief gap behind each one is rarely named, so the content plan cannot target it. A community specialist new to the data and an academic expert get the same deck.

DAY 2 MORNING · WED OCT 14

2nd of 4 to present

01 **02** 03 04

About 20 min, including a few questions. Order 01→04, after the final build.

THE PROBLEM PROMPT

> Build an AI workforce that reads field observations, advisory-board notes and enquiries, raises safety findings first, and traces each care gap to the HCP knowledge or belief gap behind it (source IDs, confidence). If evidence exists, it matches approved content to each HCP segment and stage (not aware, aware, learning, applying); if not, it hands the evidence gap off. Humans approve the gaps and sign off the brief. Stretch: a dashboard of Next Best Action by segment and gap closure over time.

Teams can keep this prompt or narrow it to one workflow they know. A good narrowed prompt names the role, the trigger, the deliverable and the decision it serves.

DONE LOOKS LIKE · DAY 1

The swarm ran end to end at least once on practice data; it stopped at Gate 1 and Gate 2 and the team made both decisions; screenshots and minute-level timings are in the capture log; the agent has drafted a slide outline. Stretch: a first dashboard, if time allows.

DONE LOOKS LIKE · DAY 2

The deck is built from the final template: no [brackets] left, impact numbers are the team's own, practice data marked fictional, the LIVE DEMO slide left empty, backup screenshots in place. The team presents second of four, for about 20 minutes with a short live demo.

LEAD SKILL

`field-insight-synthesis`

SUPPORTING SKILLS

`insight-generation`

`evidence-gap-analysis`

`meeting-transcription`

DATA PACK

Any TA; synthetic data (fictional Nordvant Biopharma) with seeded safety findings and a promotional objective to catch.

`field-observations.csv` `interaction-notes.md`

`advisory-board-transcript.md` `guideline-landscape.md`

+ practice CRM

Fictional practice data (Nordvant Biopharma), or bring your own non-confidential data.

WARM-UP MISSIONS

L2 `field-insights`
What should leadership know?

L1 `medical-information`
Handle the enquiry queue

The build afternoon, block by block

Optional order, approximate durations; teams may move faster or slower. **No formal share-out at the end of the afternoon:** facilitators circulate and ask questions (page 8).

1 · Setup — about 30 min
2 · Warm-ups — about 40 min
3 · Ideation — about 45 min
4 · The hack — about 80 min
Then cocktail reception

1 Set up Grok Bot + skills

about 30 min

Huddle about 5 min
Everyone opens aiinaction.up.railway.app/grokbot about 10 min
The driver pastes the setup prompt below about 10 min
Smoke test passes about 5 min

DONE LOOKS LIKE

Every member signed in; the driver's conversation has the skills loaded and has read one practice file; capture log started

FACILITATORS CIRCULATE

Check every laptop is signed in and the smoke test shows a file marked SYNTHETIC. GitHub blocked? Point to the starter bundle on /grokbot.

2 Warm-up missions

about 40 min

Split into two pairs about 5 min
Run both missions in parallel about 20 min
Push on it about 10 min
Swap outputs about 5 min

DONE LOOKS LIKE

Field insights (safety finding raised first) and an enquiry register exist; the team agreed one shared gap; 3 observations logged

FACILITATORS CIRCULATE

Let the missions finish without steering. Watch for the safety stop and the human-review moment; keep the 2-minute swap on time.

3 Workflow ideation

about 45 min

Map today's workflow for one real gap-to-content cycle about 15 min
Spot the AI ideas about 10 min
Pick one problem about 10 min
Sketch the swarm on paper about 10 min

DONE LOOKS LIKE

The gap-to-content cycle mapped; one problem chosen; hours per cycle and number of people involved estimated; ideas marked Efficiency or Opportunity

FACILITATORS CIRCULATE

Ask the ideation questions (page 8). Push for one problem statement and a baseline estimate before they sketch the swarm.

4 The hack

about 80 min

Write the assignment about 10 min
Run 1 about 35 min
Optional stretch break about 5 min
Run 2 about 20 min
Capture about 10 min

DONE LOOKS LIKE

Gap list (knowledge or belief → behaviour → care) with source IDs and an evidence route; segment × stage content picks and a brief, stopping at both gates; safety list first; screenshots and timings captured

FACILITATORS CIRCULATE

Ask the hack, gate and data questions. Near the end of the afternoon, check that screenshots and the capture log are saved.

DAY 2 · WED OCT 14 · FINAL BUILD → PRESENTATIONS IN ORDER 01→04

Final build + rehearsal about 30–40 min
re-open → last fix, agent fills the template → team review → rehearse → MC opens

01 Publications Brain about 20 min
02 Insights Engine about 20 min
03 Congress Monitor about 20 min
04 Field Intelligence Engine about 20 min
Then WPP demo

Insights Engine presents second of four, about 20 minutes, give or take, including a short live demo and a few questions. Keep all decks on one laptop.

Warm up, then pick your data

Warm-up missions

Block 2 · about 40 min

Level 2 · Pair — **Pair A**

"What should leadership know?"

Field observations to insights. Watch for the safety finding first, then ask which HCP knowledge or belief gap sits behind each care gap.

aiinaction.up.railway.app/missions/**field-insights**

Level 1 · Starter — **Pair B**

"Handle the enquiry queue"

The enquiry queue is a second gap signal: what HCPs keep asking tells you what they do not yet know.

aiinaction.up.railway.app/missions/**medical-information**

STRETCH, ALTERNATIVES AND REFERENCES

L1 Starter · L2 Pair · L3 Team · L4 Swarm

L1 Alternative
Turn the call notes into useful follow-up
/missions/msl-post-call

L3 Hack reference
Work across a practice CRM and content library
/missions/connected-planning

L3 Optional stretch
Run the next 30 days of Medical Affairs
/missions/thirty-day-capstone

L4 Also useful
Build the whole medical launch plan
/missions/launch-plan-swarm

BG Background
Team mission 2: the original team mission
Medical-Affairs-Skills · workshop

Data pack

browse and copy links at aiinaction.up.railway.app/data

Fictional practice data, or bring your own non-confidential data.

Nordvant Biopharma and its products are fictional. Own data: no patient data, no confidential company data, no real HCP personal data.

PICK ONE TA — MM · NORVANTIB · AD · DERMALYX · Obesity · ADIPOSYN

Any TA; synthetic data (fictional Nordvant Biopharma) with seeded safety findings and a promotional objective to catch.

TA PACK FILES synthetic/<TA>/

- TA field-observations.csv
- TA interaction-notes.md
- TA advisory-board-transcript.md
- TA guideline-landscape.md
- TA medical-information-enquiries.csv
- TA evidence-landscape.md
- TA medical-education-needs.md
- TA medical-plan.md
- TA mlr-review-comments.md

PRACTICE CRM synthetic/connected/

- CRM hcps.csv
- CRM interactions.csv
- CRM engagement.csv
- CRM content_assets.csv
- CRM enquiries.csv
- CRM data-dictionary.json

Fictional Nordvant data: hcps.csv for segments, content_assets.csv as the content list, interactions.csv and engagement.csv for the dashboard. Keep source IDs on every gap.

Efficiency vs Opportunity, and the digital team

Efficiency AI today's work, faster

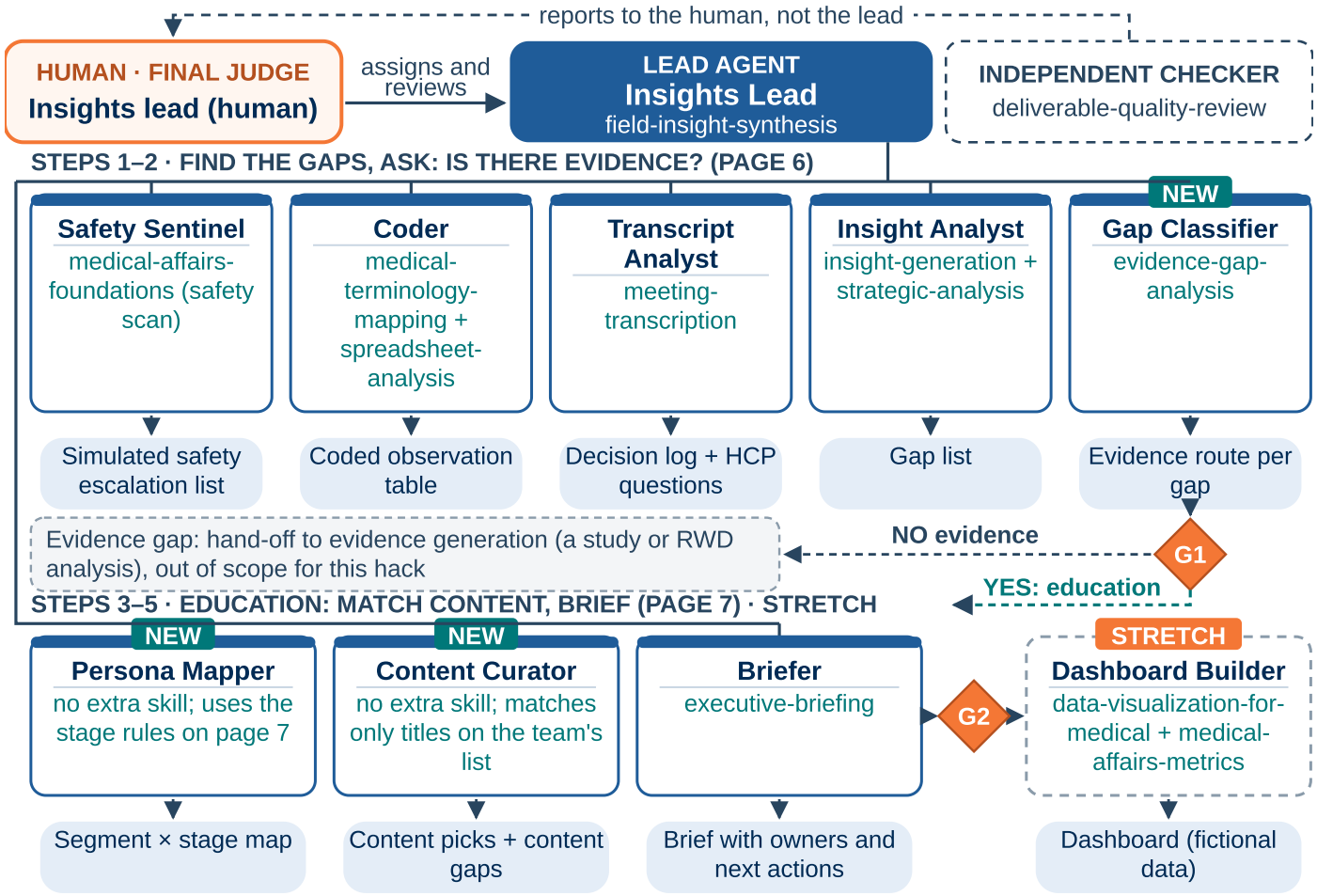
- **E1** Gap tracing: every coded observation traced from HCP knowledge or belief gap to behaviour and care gap, with its source ID and quoted words.
- **E2** First draft of a gap list (gap, chain, evidence route, source IDs, signal strength, segment, stage) ready for the insights lead to edit.
- **E3** Advisory-board transcript to a list of the questions and doubts HCPs raised, each quoted with the speaker, the same afternoon.

vs

Opportunity AI something not possible today

- **01** Content matched to each HCP: for every gap, a short list of approved content for each segment and stage (not aware to applying).
- **02** Gaps with no content: a running list of gaps that nothing in the library covers yet, passed to the medical content and education team.
- **03** One gap, every source: the same gap in MSL notes, advisory boards and congress side by side, plus what the field did not raise.

Suggested swarm one lead agent, sub-agents with their skills, two human checks



Dashed arrow = hand-off · orange diamond = a human approves before work goes on. G1: the insights lead approves the top 3–5 gaps, their evidence route and the safety routing. G2: the head of medical signs off the brief.

Find the gaps, then follow the evidence

Steps 1–2 of five, for the hack (Block 4, about 80 min). **Top 3–5 gaps** only: the agent drafts, the insights lead decides what counts as a gap. Steps 3–5: page 7.

1 Gather the insights

Safety Sentinel · Coder · Transcript Analyst

- **MSL field insights** `field-observations.csv` `interaction-notes.md`
- **Advisory boards** `advisory-board-transcript.md`
- **Enquiries** `medical-information-enquiries.csv`
- **Congress** own non-confidential notes; `evidence-landscape.md` kept separate

Safety scan first. Possible AEs, product complaints and off-label mentions listed verbatim and routed before any gap work. Output: coded table with source IDs.

Signal strength, then pick the top 3–5

Every gap carries its **source IDs**, quoted words, **record count**, **number of sources** and a confidence. Flag weak signal when it rests on one or two records or one source; weak signals keep their own section, not dropped, not ranked.

> **G1:** the insights lead approves the top 3–5 gaps, their evidence route (hand-offs too) and the safety routing.

2 Name the gap: one chain

Insight Analyst

A care gap **always** stems from a knowledge or belief gap:

HCP knowledge or belief gap → **HCP behaviour gap** → **Patient care gap**

Knowledge gap: does not know or misreads the evidence.
Belief gap: knows it but doubts it, or doubts it applies to their patients.

Irrespective of guidelines: even when it is in the guideline, knowledge and perception may not have shifted.

Suspected care gap: every care gap from field reports is "suspected": an MSL's account is not practice data; name the data that would confirm it.

2 Then ask: is there evidence?

Gap Classifier

Is there evidence? guidelines, publications

YES Education
Medical content creation + medical education, steps 3–5. This vertical's focus, led by John and Anjali.

NO Evidence gap
Evidence generation (a study or RWD analysis): a hand-off, out of scope for this hack. List it and stop.

THREE EXAMPLE ROWS EXAMPLE

illustrative only, not from the data pack

Record (source)	Chain: knowledge or belief → behaviour → care	Signal	Evidence? → route · segment · stage
"Asked how often to check labs after switching therapy." (MSL note)	knowledge → labs not checked on time → suspected care gap	[n] records, [n] sources	yes (label, guideline) → education · community specialist · applying
"Most patients here start treatment without the recommended test; I'm not sure the result changes my choice." (MSL note)	belief → test skipped → suspected care gap	1 record, 1 source weak signal	yes (guideline) → education · community specialist · learning
"Is there anything on patients over 75? I have nothing to go on." (advisory board)	knowledge → no shared practice → suspected care gap	[n] records, [n] sources	no → evidence gap · hand-off · academic physician

OUTPUT OF STEPS 1–2: THE GAP LIST, ONE ROW PER GAP

```
Gap | Chain | Suspected? | Evidence? | Route | Source IDs | Records | Sources | Weak signal? | Confidence
```

Match content to each HCP segment

Steps 3–5, approved education gaps only. **The Scribe first pastes the team's content list**: `Title | type | stage | proactive or reactive only | status`.

3 Map to segment and stage — Persona Mapper

Personas are **segments**, not profiles of named HCPs. Name three to five, e.g. community specialist · academic physician · primary care · NP or PA · pharmacist. No role or setting in the record: **unknown**; do not guess.

Stage	Signs in the record
Not aware	Does not mention the evidence or topic at all; describes practice without it; is surprised when the MSL raises it
Aware	Has heard of it but has no detail. Asks "what is it?" or "does this apply to my patients?"
Learning	Weighing the data: asks about study design, comparisons, subgroups, which patients fit
Applying	Has used it or plans to: asks about dosing, monitoring, side effects, switching, hard cases

Two stages? Tag the earlier. Give a **record count** per cell.

4 Pick content for each cell — Content Curator — types and channels are EXAMPLES

Stage	Content type	Channel	Title	Use	Status
Not aware	short evidence summary, one-page explainer	MSL visit; congress medical booth	from list or **to match**	proactive / reactive only	from list, else **check before use**
Aware	plain-language overview, key data summary	MSL visit; MSL follow-up email	as above		
Learning	publication summary, data slides, congress highlights	MSL scientific exchange; webinar or peer meeting	as above		
Applying	practical guide or checklist, case discussion	MSL follow-up; medical information response	as above		

- Use and status are copied, never judged; missing status = **check before use**. Off-label: reactive only, via **medical information**.
- No title fits: **to match**, added to the content gap list for medical content and education. Never write new content.

5 Brief and sign-off — Briefer · checker

Each gap: **owner** + **next action**; evidence gaps listed for hand-off. **G2:** head of medical signs off.

OUTPUTS AT THE END OF THE HACK

Gap list with evidence routes · segment × stage map with counts · content picks + content gap list · evidence gaps for hand-off · draft brief · capture log

STRETCH GOAL The dashboard — Dashboard Builder, optional

After G2: **Next Best Action by HCP segment**, **knowledge-gap closure over time**, or both.

Next Best Action by segment



next content + channel per segment × stage, HCPs it applies to

Gap closure over time



share of HCPs per stage, by month (not aware → applying)

FICTIONAL Nordvant data (interactions, engagement); counts, not causal claims. Sketch only.

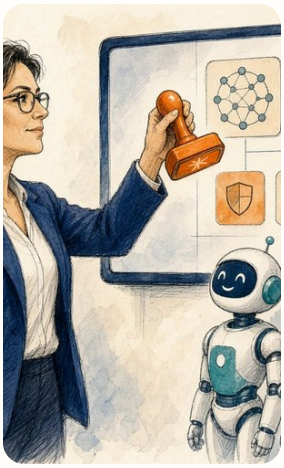
Where humans decide, and what to ask

G1 Gate 1

WHEN	After the Gap Classifier
WHO DECIDES	**The insights lead**
WHAT THEY CHECK	Approves the top 3–5 gaps, their evidence route and the safety routing before any content is matched.
WHY A HUMAN	What counts as a gap is a judgment, and safety routing is a regulated obligation.

G2 Gate 2

WHEN	After the Briefer
WHO DECIDES	**The head of medical**
WHAT THEY CHECK	Signs off the brief, the owners and the next actions before it is shared beyond the team.
WHY A HUMAN	Accountability for decisions and for what HCPs are told.



Always-on stop (a rule, not a gate): any possible adverse event, product quality complaint or off-label signal in human-sourced records is listed verbatim and routed by a human before analysis continues.

Facilitator question bank

the Day 1 feedback loop: ask one or two per visit, then move on

Ideation block 3

1. Where do care gaps surface in your team today: field insights, advisory boards, medical information? Who traces them back to what HCPs know or believe?
2. Pick one care gap. Which HCP knowledge or belief gap sits behind it, and which HCP behaviour does it drive?
3. When is a gap an education gap, and when is it an evidence gap? Who owns each one today?
4. Which HCP segments matter most for this TA, and how would you tell which stage (not aware to applying) each one is at?

The hack block 4

5. Pick one segment × stage cell. Why that content and channel? Is every title from your list, with its status copied rather than judged?

Data during the hack

6. Did your swarm raise the seeded safety finding before anything else? Who receives it, and what does the agent stop doing?
7. Show me one gap and the record IDs behind it. Can you read the chain from knowledge or belief gap to behaviour to care gap?
8. This gap is in the guideline, yet HCPs still do not act. Did the Gap Classifier send it to education rather than evidence generation? Why?
9. Did you try the stretch dashboard? What is the Next Best Action for one segment, and what would show a gap closing over time?

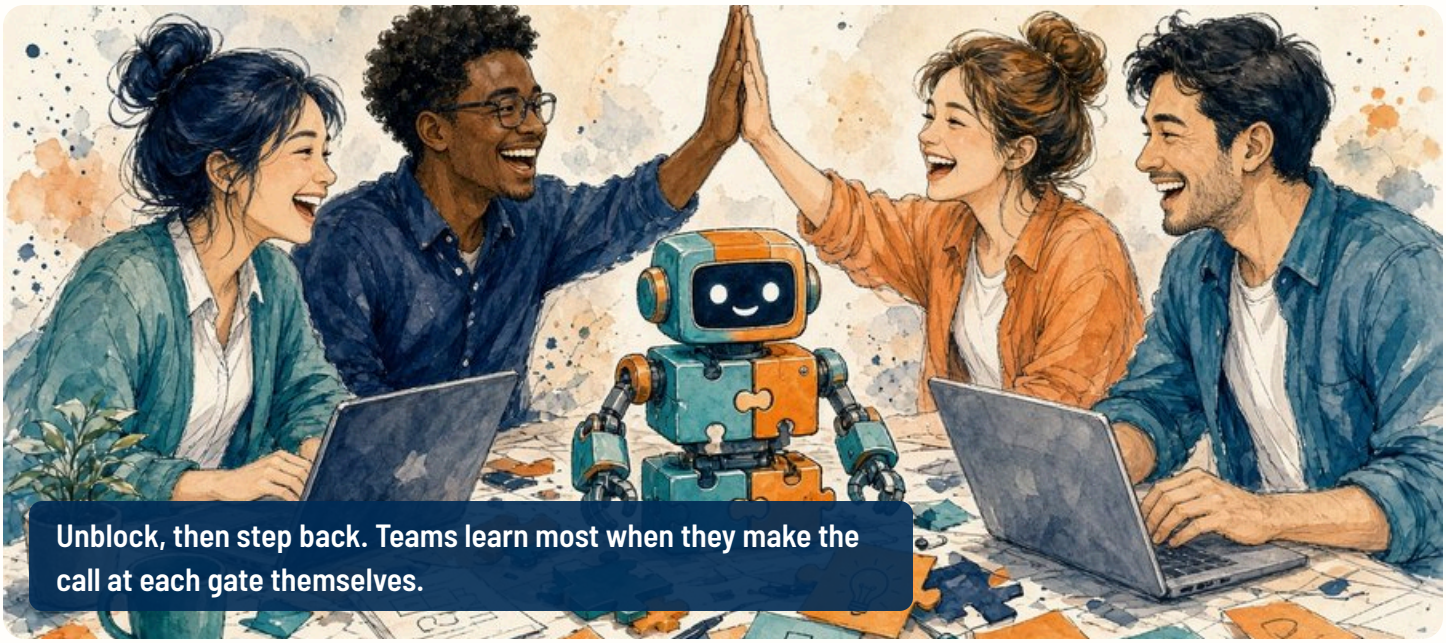
Impact during the hack

10. How many hours does one gap-to-content cycle take today, across everyone involved? What does the AI-assisted cycle take, including review?

When a team gets stuck

The ten snags facilitators see most often in this vertical, each with the unblocker to offer. Nudge, then step back: the team does the building.

Pitfall	How to unblock
01 Care gaps with no cause named.	Ask: "Which HCP knowledge or belief gap sits behind it, and which behaviour does it drive?" Trace the chain first.
02 "It is in the guideline, so it is not a gap."	Gaps are defined by what HCPs know and believe, irrespective of guidelines: guidance alone does not shift perception.
03 Every gap becomes an evidence gap.	The Gap Classifier checks guideline-landscape.md and evidence-landscape.md first; real evidence gaps are a hand-off, out of scope today.
04 The safety scan is skipped or buried.	Make the Safety Sentinel the first worker and tell the lead to stop until a human acknowledges the list.
05 Segments made up rather than drawn from the data.	Build segments from hcps.csv (setting); if role or setting is missing, the segment is unknown.
06 The Curator writes new content or judges approval.	Titles only from the pasted list, status copied; no fit = "to match" on the content gap list.
07 Promotional drift.	Keep objectives about knowledge and practice, not product use; off-label stays reactive, routed to medical information.
08 A run is slow or wanders.	Plan first, one sub-agent at a time; top 3–5 gaps, one TA, two segments; the dashboard only in the second run.
09 Grok Bot cannot open GitHub.	Upload the starter bundle for your TA from https://aiinaction.up.railway.app/grokbot, or the repository ZIP.
10 The output is generic.	Teach the agent: add 3–5 house rules (what good looks like, what must never appear) and run again.



Unblock, then step back. Teams learn most when they make the call at each gate themselves.

Impact worksheet and setup prompt

Impact worksheet the team's own estimates

One cycle = one gap-to-content cycle (gaps found, routed and matched to content for each segment and stage)

Today baseline	h	×		×	$ /h	=	$
	hours per cycle (all people)		cycles per year		loaded rate		cost per year
			vs				
AI-assisted with the swarm	h	×		×	$ /h	=	$
	hours per cycle, incl. human review		cycles per year		loaded rate		cost per year

HOURS SAVED PER YEAR

(today h − AI h) × cycles = h

VALUE PER YEAR (ESTIMATE)

hours saved × rate = $

OTHER BENEFITS

days	%	#
speed to decision	coverage	errors caught

ILLUSTRATIVE **Not data, only an example:** 60 h → 12 h per cycle × 4 cycles/year = 192 h saved; × $150/h = $28,800 per year. Replace every number with the team's own estimate.

Count every person involved and include human review time in the AI-assisted row. These numbers go on the Impact slide of the final template (slide 9: formula tiles and a bar chart built from the team's own estimates).

Setup prompt Block 1, after sign-in · the driver pastes it into Grok Bot (replace the TA)

1 · Setup prompt copy all

```
Use https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills
Read AGENTS.md. We are the Insights Engine team at AI in Action for Medical Affairs.
Load medical-affairs-foundations and these skills: field-insight-synthesis, insight-generation, strategic-analysis, evidence-gap-analysis, meeting-transcription, medical-terminology-mapping, executive-briefing, deliverable-quality-review.
Use the practice data in https://github.com/Open-Medical-Affairs/Data-Sources: synthetic/[oncology-mm | immunology-ad | cardiometabolic-obesity]/ and synthetic/connected/ (the practice CRM). It is fictional (Nordvant Biopharma).
First, only: list the skills you loaded, then show the first 5 rows of field-observations.csv and confirm the file is labelled SYNTHETIC. Do not start a mission yet.
Do not ask me to connect company systems. Do not send, post or change anything outside this conversation.
```

SUGGESTED ROLES (ROTATE IF YOU LIKE)

- **Driver** types into Grok Bot
- **Navigator** reads the mission and pushes back
- **Scribe** owns the capture log
- **Timekeeper + presenter** keeps the clock, tells the story

02 COPYABLE PROMPT

Assignment starter for the hack

Paste into Grok Bot at the start of Block 4 (the hack), then edit the [brackets]. The setup prompt for Block 1 is on page 10. Ready prompts for every mission: **aiinaction.up.railway.app/prompts** · swarm mode and the Skill creator: **/optimizer**.

2 · Assignment starter: Insights Engine swarm — **Block 4 · start of the hack**

```
# Assignment: Insights Engine swarm
You are the Insights Lead, coordinating digital workers for a Medical Affairs team. This is an assignment:
plan, delegate, check, hand back drafts.
## Goal: what done looks like
[One sentence: our problem and the deliverable.]
## Context to use
- https://github.com/Open-Medical-Affairs/Medical-Affairs-Skills : read AGENTS.md; load medical-affairs-
foundations, field-insight-synthesis and the skills below.
- Practice data (fictional Nordvant Biopharma): https://github.com/Open-Medical-Affairs/Data-Sources
synthetic/[TA]/ and synthetic/connected/. Or our own non-confidential data: [describe].
- Our HCP segments: [3-5]. Our content list, one line each: Title | type | stage | proactive or reactive only
| status: [paste].
## How to work
1. Show me a short plan and the org chart first. Wait for my OK.
2. Safety first: list possible AEs, product complaints and off-label use verbatim; stop until I acknowledge.
3. Run these sub-agents, each with only the context it needs:
   - Safety Sentinel (medical-affairs-foundations): the step-2 scan. Hands off: Simulated safety escalation
list.
   - Coder (medical-terminology-mapping + spreadsheet-analysis): codes observations, keeps source IDs. Hands
off: Coded observation table.
   - Transcript Analyst (meeting-transcription): HCP questions and doubts, quoted with speaker. Hands off:
Decision log + HCP questions.
   - Insight Analyst (insight-generation + strategic-analysis): name each chain: HCP knowledge/belief gap ->
behaviour gap -> patient care gap (suspected if only from field reports), judged by what HCPs know and
believe, even if a guideline covers it. Record and source counts; flag weak signals (1-2 records or one
source); top 3-5. Hands off: Gap list.
   - Gap Classifier (evidence-gap-analysis): evidence exists (guidelines, publications)? Yes: education. No:
evidence gap, hand-off to evidence generation (study or RWD); stop there. Hands off: Evidence route per gap.
   - Persona Mapper (no extra skill): segment (from our list, or unknown) and stage (not aware, aware,
learning, applying; if two, the earlier) per education gap, with record counts. Hands off: Segment x stage
map.
   - Content Curator (no extra skill): per cell, type, channel and titles only from our list, else "to match";
copy use and status (missing: "check before use"). Off-label: reactive only, via medical information. Never
write new content. Hands off: Content picks + content gap list.
   - Briefer (executive-briefing): owner and next action per gap; evidence gaps listed separately. Hands off:
Brief with owners and next actions.
4. STOP at Gate 1: After the Gap Classifier: the insights lead approves the top 3-5 gaps, their evidence route
and the safety routing before any content is matched. Wait.
5. An independent checker (deliverable-quality-review, reporting to me) reviews the draft; fix what it finds.
6. STOP at Gate 2: After the Briefer: the head of medical signs off the brief, owners and next actions. Wait.
7. Stretch, only after Gate 2: Dashboard Builder (data-visualization-for-medical + medical-affairs-metrics):
Next Best Action by HCP segment and/or knowledge-gap closure over time. Label it FICTIONAL. Hands off:
Dashboard.
8. Keep a timing note: minutes per step and per gate.
## House rules
- Never invent a citation, number or quote; mark unsupported claims. Keep source IDs; keep public evidence
separate.
- Do not send, post, publish or change any external system. Drafts only. Mark every output DRAFT and the data
FICTIONAL.
```

All products, people and results in the practice data are fictional. Outputs are drafts that need qualified human review; this is not a clinical decision, GxP or pharmacovigilance system.